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/*LAB-2 Program to find generators and corresponding possible subgroups of a cyclic group*
kill(all)$
G:{1,-1,%i,-%i};
flag:0$
n:length(G)$
m:0$
for a in G do (
S:{},
for j:1 thru n do (
S:adjoin(a^j,S)),
print("subgroup generated by", a, "is", S),
if G=S then
block(
print(a," is a generator"),
m:m+1, flag:1 ) )$
if flag=0 then
print("G has no generator and not cyclic group")
else
print("It is cyclic group and number of generators",m)$
    {-1,1,-%i,%i}
subgroup generated by -1 is {-1,1}
subgroup generated by 1 is {1}
subgroup generated by -%i is {-1,1,-%i,%i}
-%i is a generator
subgroup generated by %i is {-1,1,-%i,%i}
%i is a generator
It is cyclic group and number of generators 2

```

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/*LAB-1 Program to construct Cayley table and test abelian for given finite set. */
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```
kill(all)$
S:listify({0,1,2,3,4,5})$
bo(a,b):=mod(a+b,6)$
k:length(S)$
S1: makelist(makelist(bo(S[i], S[j]), j, k), i, k)$
print("Cayley Table is")$
S2: apply(matrix, S1);
if S2=transpose(S2) then
print("given binary operation is abelian")
else
print("given binary operation is not abelian")$
```

Cayley Table is

0	1	2	3	4	5
1	2	3	4	5	0
2	3	4	5	0	1
3	4	5	0	1	2
4	5	0	1	2	3
5	0	1	2	3	4

given binary operation is abelian

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/* LAB-3 Verify Lagrange's theorem for H={1,-1} of a group G={1,-1,i,-i} */
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```
kill(all)$
G:{1,-1,%i,-%i};
H:{1,-1};
print("Order of G =",n:length(G))$
print("Order of H =",m:length(H))$
H1:cartesian_product(H,H)$
H2:makeset(a.b,[a,b],H1)$
if H2=H then
  if mod(n,m)=0 then
    disp("H is a subgroup of G and Lagrange's theorem is satisfied")
  else
    disp("Lagrange's theorem is not applicable")
else
  disp("H is not a subgroup of G")$
```

$\{-1, 1, -i, i\}$ $\{-1, 1\}$

Order of G = 4

Order of H = 2

H is a subgroup of G and Lagrange's theorem is satisfied

/* S:{1,2,3,4,5}; for finite set, S:[2,3]; for interval, S:(2*n+4)/(3*n+6); infinite set
type:finite; for finite set, type:interval; for interval, type:infinite; infinite set */

/*LAB-4 Program to find supremum and infimum of set */

```
kill(all)$
S:(n+1)/(n+2);
type:infinite$
if type=interval or type=finite then
block(
S:listify(S),
print("sup(S) = ",lmax(S)),
print("inf(S) = ",lmin(S))
)
else
block(
L:limit(S,n,1),
M:limit(S,n,inf),
print("sup(S) = ",max(L,M)),
print("inf(S) = ",min(L,M))
)$
```

$$\frac{n+1}{n+2}$$

$$\sup(S) = 1$$

$$\inf(S) = \frac{2}{3}$$

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/*LAB-5 find limit by Cauchy's first theorem on limits*/

```
kill(all)$
load('simplify_sum)$
un:n^(1/n);
L:'limit(un,n,inf)$
L1:ev(L,nouns)$
xn:sum(un,n,1,n)$
M:'limit(xn/n,n,inf)$
print(L," = ",L1)$
print("By Cauchy's first theorem on limits ")$
print(M," = ",L1)$
```

$$\lim_{n \rightarrow \infty} n^{1/n} = 1$$

By Cauchy's first theorem on limits

$$\lim_{n \rightarrow \infty} \frac{\sum_{n=1}^n \left(n^{1/n} \right)}{n} = 1$$

/*LAB-5 find limit by Cauchy's second theorem on limits*/

```
kill(all)$
ratfrac:true$
u(n):=(n^n)/(n!);
L:'limit(u(n+1)/u(n),n,inf)$
L1:ev(L,nouns)$
M:'limit(u(n)^(1/n),n,inf)$
print(L," = ",L1)$
print("By Cauchy's second theorem on limits ")$
print(M," = ",L1)$
```

$$u(n) := \frac{n^n}{n!}$$

$$\lim_{n \rightarrow \infty} \frac{(n+1)^{n+1}}{n^n (n+1)!} = e$$

By Cauchy's second theorem on limits

$$\lim_{n \rightarrow \infty} \left(\frac{n^n}{n!} \right)^{1/n} = e$$

```

/*LAB-6 Geometric series*/
kill(all)$
a:2$
r:1/3$
un:a*r^n;
s:sum(un,n,0,inf);
if abs(r)<1 then
print("The given series converges to",a/(1-r))
else
print("The given series is diverges")$

```

$$\frac{2}{3^n} \quad 2 \sum_{n=0}^{\infty} \left(\frac{1}{3^n} \right)$$

The given series converges to 3

```

/*LAB-7 P-series*/
kill(all)$
un:1/n^(1/2)$
s:sum(un,n,1,inf);
L:hipow(denom(un),n);
if L<1 or L=1 then
print("By P-series test the series is divergent")
else
print("By P-series test the series convergent")$

```

$$\sum_{n=1}^{\infty} \left(\frac{1}{\sqrt{n}} \right) \quad \frac{1}{2}$$

By P-series test the series is divergent

```

/*LAB-8 D'Alembert's ratio test*/
kill(all)$
u(n):=n!/(n^n)$
s:sum(u(n),n,1, inf);
D:limit(u(n+1)/u(n),n,inf);
if D<1 then
disp("By D'Alembert's ratio test the series is convergent")
elseif D>1 then
disp("By D'Alembert's ratio test the series is divergent")
else
disp("D'Alembert's ratio test fails ")$

```

$$\sum_{n=1}^{\infty} \left(\frac{n!}{n^n} \right) \quad \%e^{-1}$$

By D'Alembert's ratio test the series is convergent

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```

/*LAB-9 Raabe's test*/
kill(all)$
u(n):=n/(n+1);
s:sum(u(n),n,1, inf);
R:limit(n*(u(n)/u(n+1)-1),n,inf);
if R > 1 then
disp("By Raabe's test the series is convergent")
elseif R < 1 then
disp("By Raabe's test the series is divergent")
else
disp("Raabe's test fails")$

```

$$u(n) := \frac{n}{n+1} \quad \sum_{n=1}^{\infty} \left(\frac{n}{n+1} \right) \quad 0$$

By Raabe's test the series is divergent

```

/*LAB-10 Cauchy's root test*/
kill(all)$
assume (x>0)$
un:n^n$
s:sum(un,n,1,inf);
L:limit(un^(1/n), n,inf);
if L<1 then
disp("By Cauchy's root test the series is convergent")
elseif L>1 then
disp("By Cauchy's root test the series is divergent")
else
disp("Cauchy's root test fails ")$

```

$$\sum_{n=1}^{\infty} (n^n) \quad \infty$$

By Cauchy's root test the series is divergent

```

/*LAB-11 Cauchy's integral test*/
kill(all)$
assume (t>1)$
un:4/(n+1)$
s:sum(un,n,1,inf);
C:integrate(un,n,1,t);
L:limit(C,t,inf);
if abs(L)#inf then
disp("By Cauchy's integral test the series is convergent")
else
disp("By Cauchy's integral test the series is divergent")$

```

$$4 \sum_{n=1}^{\infty} \left(\frac{1}{n+1} \right) \quad 4 (\log(t+1) - \log(2)) \quad \infty$$

By Cauchy's integral test the series is divergent

```

/*LAB-12 Alternating Series*/
kill(all)$
u(n):=1/n;
s:sum((-1)^(n)·u(n),n,1,inf);
L:limit(u(n),n,inf);
if L=0 and u(3) < u(2) then
disp("By Leibnitz theorem is convergent")
else
disp("By Leibnitz theorem is divergent")$

```

$$u(n) := \frac{1}{n} \quad \sum_{n=1}^{\infty} \left(\frac{(-1)^n}{n} \right) \quad 0$$

By Leibnitz theorem is convergent